

Pre-Congress Workshop WS-13

Learning from Data: Principles, Methods, and Emerging Ideas & Applications in Iterative Learning Control

2026 IFAC World Congress
Busan, Republic of Korea
23 August 2026

Organizers:

- Ying Tan (University of Melbourne, Australia, yingt@unimelb.edu.au)
- Tom Oomen (Eindhoven University of Technology, The Netherlands, T.A.E.Oomen@tue.nl)
- Kevin L. Moore (Colorado School of Mines, USA, kmoore@mines.edu)
- Bing Chu (University of Southampton, UK, b.chu@soton.ac.uk)
- Kira Barton (University of Michigan, USA, bartonkl@umich.edu)
- Leontine Aarnoudse (Norwegian University of Science and Technology, Norway, leontine.i.m.aarnoudse@ntnu.no)



Schedule

<https://sites.google.com/view/ifac2026ilc>



Slides

<https://toomen.eu/ilc2026/>

Schedule and Speakers

Time	Speaker	Topic
09:00-09:05	Ying Tan	Welcome and Workshop Overview
09:05-09:50	Tom Oomen	Introduction to ILC and Frequency-Domain Design
09:50-10:20	Kevin Moore	Classical ILC/DRP Design from an Algebraic (Matrix-Fraction) Perspective – Part 1: Framework
10:20-10:40		Coffee Break
10:40-11:10	Kevin Moore	Classical ILC/DRP Design from an Algebraic (Matrix-Fraction) Perspective – Part 2: Design
11:10-12:00	Tom Oomen	Norm-Optimal ILC
12:00-13:30		Lunch
13:30-14:20	Ying Tan	Learnability and Persistence of Excitation
14:20-14:50	Leontine Aarnoudse	Nonlinear ILC for Disturbance Discrimination
14:50-15:20	Ying Tan	Fast-convergence ILC
15:20-15:40		Coffee Break
15:40-16:10	Kevin Moore	Data-Driven ILC, Norm-Optimal ILC, and DRP
16:10-16:40	Tom Oomen	Data-Driven Estimation
16:40-17:00	Ying Tan (moderator)	Open Discussion and Concluding Remarks



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Open Discussion

Moderator: Ying Tan

Panel: Ying Tan (University of Melbourne, Australia)
Tom Oomen (Eindhoven University of Technology, The Netherlands)
Kevin L. Moore (Colorado School of Mines, USA)
Leontine Aarnoudse (Norwegian University of Science and Technology, Norway)

Questions:

- *Mutatis mutandis*, if a new PhD student or researcher wants to work in this field, what do you suggest they work on to make the biggest impact in theory, application, or both?
- Are there any truly outstanding research questions left in multi-pass systems theory? Or, are there fundamental limits to performance in ILC that people don't seem to understand?
- Will AI render ILC and DRP obsolete? Or, how can AI be used in the field?
- How can ILC be truly data-driven when so many results depend on models? Or, what is the role of models in data-driven ILC and DRP?



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Thank you for attending!



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